

Owen Lee

312-296-4285 | owenlee3@illinois.edu | github.com/Owen2215 | linkedin.com/in/owenlee3

EDUCATION

The University of Illinois Urbana-Champaign <i>Master of Computer Science</i>	Urbana, IL <i>Expected May 2028</i>
The Chinese University of Hong Kong, Shenzhen <i>Bachelor of Engineering, Computer Science and Engineering</i> <i>Relevant Coursework: Natural Language Processing, Deep Learning, Machine Learning, Operating Systems</i>	Shenzhen, China <i>May 2026</i>

SKILLS

Programming Languages: Python, Java, C++, Javascript, HTML, SQL
AI/ML: Large Language Models, Retrieval-Augmented Generation (RAG), Fine-tuning (LoRA/QLoRA), Natural Language Processing, Generative AI
ML Frameworks: PyTorch, HuggingFace (Transformers, PEFT), NumPy, scikit-learn, Vector DBs (Weaviate, ChromaDB)
Technologies & Infra: Spring Boot, REST APIs, AWS (EC2, Lambda, Bedrock), Docker, Git, Linux

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineering Intern <i>ATG Nexus Sdn Bhd</i>	July 2025 – October 2025 <i>Puchong, Malaysia</i>
- Built hybrid keyword+vector retrieval (Weaviate + OpenAI embeddings) for a production legal search app - Engineered bilingual triage report generator (Qwen3-Embedding-8B) and Gemini + Plotly Dash pipeline for automated insights, increasing analyst throughput by 50%	
Software Engineering Intern <i>China Unicom</i>	May 2024 – August 2024 <i>Zhejiang, China</i>
- Built a Java and Spring Boot REST API for real-time weather dashboards, sustaining a 95% success rate at ~2,400 req/hr over 5 weeks in a high-scale production environment - Implemented server health monitoring with 99% alert detection accuracy, validated against 4 weeks of alert logs	

RESEARCH EXPERIENCE

Research Assistant (Advisor: Prof. Xiaoying Tang) <i>The Chinese University of Hong Kong, Shenzhen</i>	November 2025 – May 2026 <i>Shenzhen, China</i>
- Engineered 1,040 step-grounded visual-text pairs for VLM training, contributing to ICML 2026 acceptance - Designed CoT prompting templates aligning rationales with rendered intermediate states, improving training signal for visual reasoning benchmarks	
Research Assistant (Advisor: Prof. Benyou Wang) <i>The Chinese University of Hong Kong, Shenzhen</i>	January 2025 – May 2025 <i>Shenzhen, China</i>
- Built multimodal crawler (1.2M audio-caption pairs) with CLAP-filtered pipeline, cutting hallucination by 15% - Trained HTSAT+BERT on EC2-scaled S3 pipeline; achieved 44.3/57.8 R@1 (text→audio / audio→text), best among 5 baselines — contributing to ACL 2026 acceptance	

PROJECTS

MoE Routing vs. Dense Finetuning vs. Zero-Shot API <i>PyTorch, HuggingFace, QLoRA</i>	Aug 2026
- Tested a router dispatching to 2B QLoRA specialists against a dense QLoRA 4B model and zero-shot Gemini, scoring 1,350 examples across 3 reasoning tasks with task-specific semantic verifiers - Both finetuned approaches beat zero-shot API calls by an order of magnitude, and the dense 4B model outperformed the routed specialists by ~30% relative accuracy despite 100% router accuracy — showing task reasoning depth, not architecture, was the real bottleneck	
Retrieval-Grounded Text-to-SQL Chatbot <i>Python, MySQL, ChromaDB, SQLGlue</i>	October 2025
- Architected schema-aware RAG pipeline (all-MiniLM + ChromaDB) for SQL generation grounded in live DB metadata - Built an execution-based evaluation harness (500+ queries) to score generated SQL against live DB metadata, measuring 89% execution accuracy - Added SQLGlue guardrails and a 3-try self-repair loop, eliminating unsafe execution	

PUBLICATIONS

Y. Huo, S. Zhang, K. Zeng, H. Liu, **O. Lee**, J. Chen, Y. Lu, Y. Guo, Y. Liang, X. Tang. *Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought*. [Paper](#). International Conference on Machine Learning (ICML), 2026.

S. Chen, X. Yuan, Z. Li, L. Zhao, **O. Lee**, Z. Su, Q. Sun, B. Wang. *FusionAudio-1.2M: Towards Fine-grained Audio Captioning with Multimodal Contextual Fusion*. [Paper](#). Association for Computational Linguistics (ACL), 2026.